

**GRAPH THEORY:
HOMEWORK 5
BOOK SECTION 3.1.**

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1. let S be the set of vertices saturated by a matching M in a graph G . Prove that some maximum matching also saturates all of S . (Hint: Use Berge's Theorem to show that one can start with any matching M , and find a sequence of "alternating path augmentations" that produce a maximum matching M^* . Be careful not to conflate "maximum" with "maximal.")

Proof. □

2. Determine the minimum size of a maximal matching in the cycle C_n ,

Solution. □

3. Let M and M' be matchings in an X, Y -bigraph G . Suppose that M saturates $S \subseteq X$ and M' saturates $T \subseteq Y$. Prove that G has a matching that saturates $S \cup T$

Proof. □

4. When $k \geq 2$, prove that Q_k has at least $2^{2^{k-2}}$ perfect matchings

Proof. □

5. Let G be the Petersen Graph.

(a) Prove that deleting any perfect matching of G leaves $C_5 + C_5$ (i.e. two disjoint 5-cycles). (Hint: After you remove a perfect matching what is the degree of each vertex).

(b) Prove that the Petersen graph has exactly six perfect matchings. (Hint: use (a) and use the fact that you already know how many 5-cycles G contains.)

Proof. □